

Gaussian Process Regression-Based

Bump Hunting

in Future Searches for the

X17 Boson

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01

The *X17* Boson

EXPERIMENTAL *EVIDENCE*

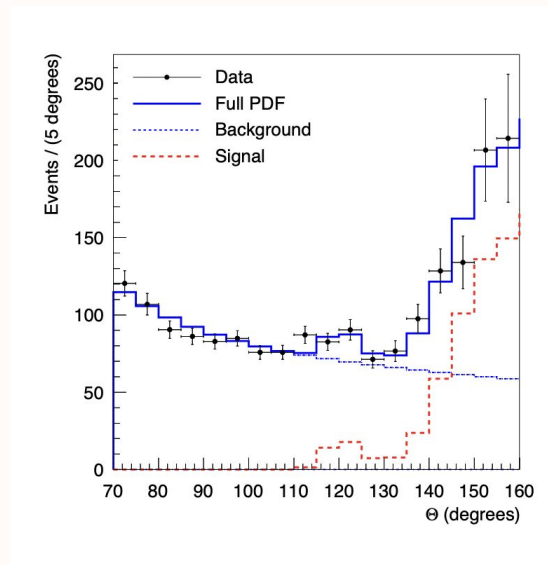
ATOMKI

ATOMKI experiments found excesses in e^+e^- pairs from nuclear transitions in ^8Be , ^4He , and ^{12}C at ~ 17 MeV.

INDEPENDENT EXPERIMENTS

Hanoi University of Sciences (Vietnam) and Joint Institute for Nuclear Research (Russia) have also found significant evidence of the X17 particle.

New experiments proposed at GANIL in France, the Pelletron accelerator in Melbourne, Jefferson Lab, and more.

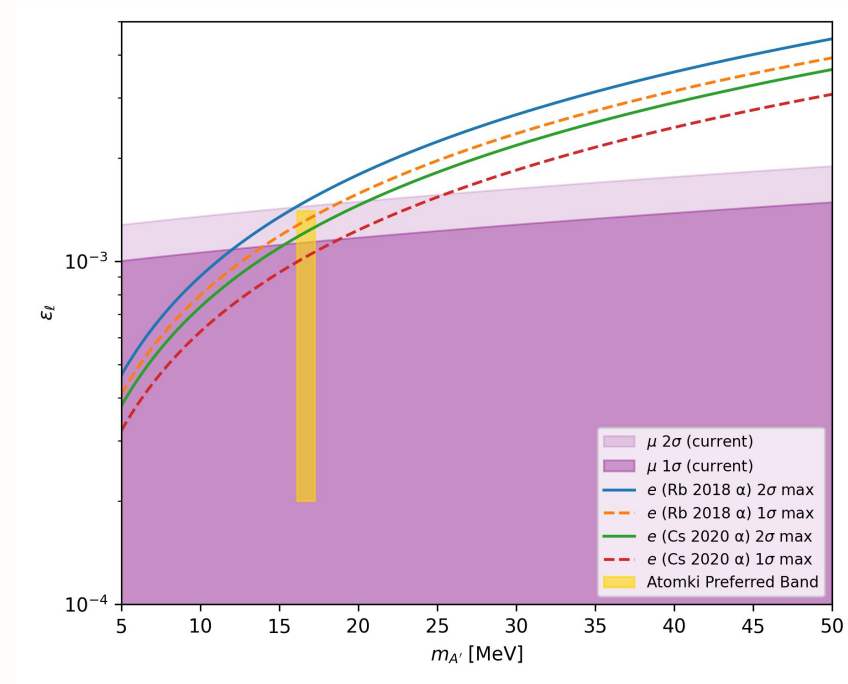


EXPERIMENTAL *EVIDENCE*

PADME

Recent results from PADME and NA64 along with NA48/2 results suggest that the X17 needs to be protophobic. The possible coupling of the X17 seems to be

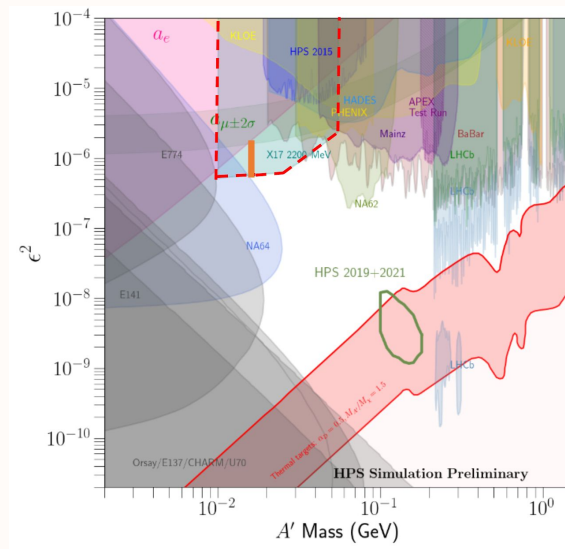
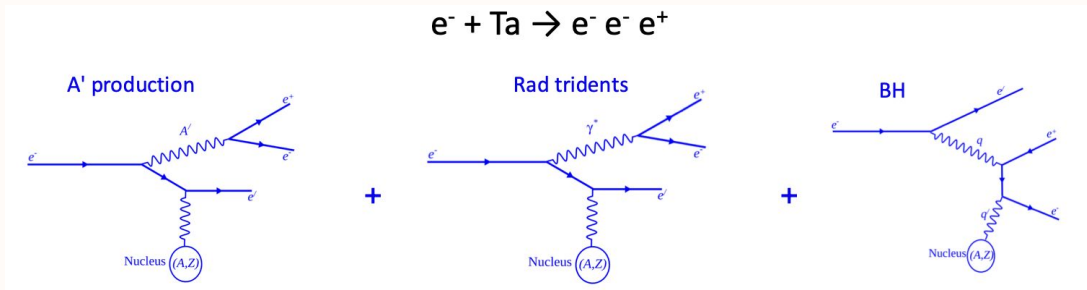
$$3.4 \times 10^{-4} \lesssim \varepsilon \lesssim 5.6 \times 10^{-4}$$



^From E.Peets on viability of protophobic explanation for X17

THE *JEFFERSON LAB* EXPERIMENT

- Uses a magnetic-spectrometer-free setup (the PRAD apparatus)
- Focusses on the bremsstrahlung-like production of hidden sector particles from electron scattering.
 - Looks for a dark photon in the mass range 3–60 MeV through it's $A' \rightarrow e^+e^-$ decay
- Scheduled to run in May 2026



Jefferson Lab

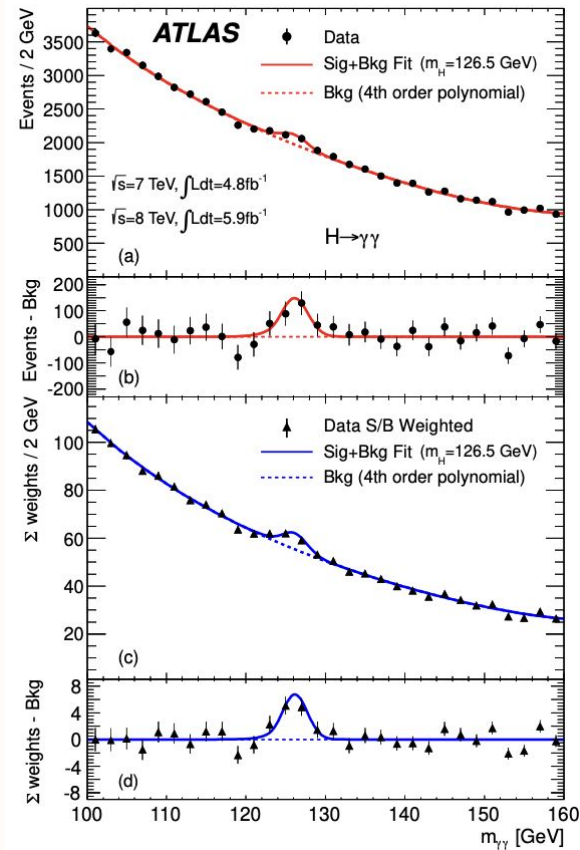
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02

Bump *Hunting*

A Bump Hunt

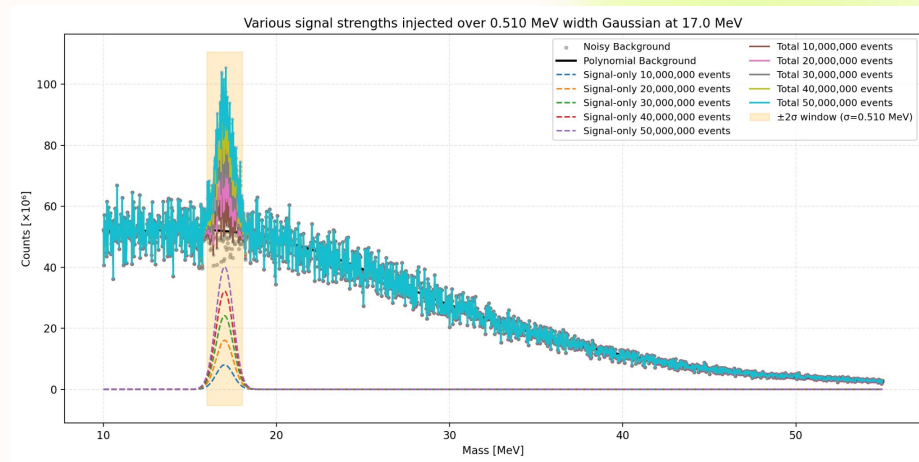
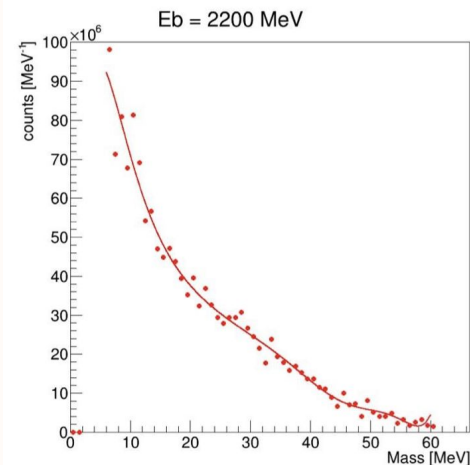
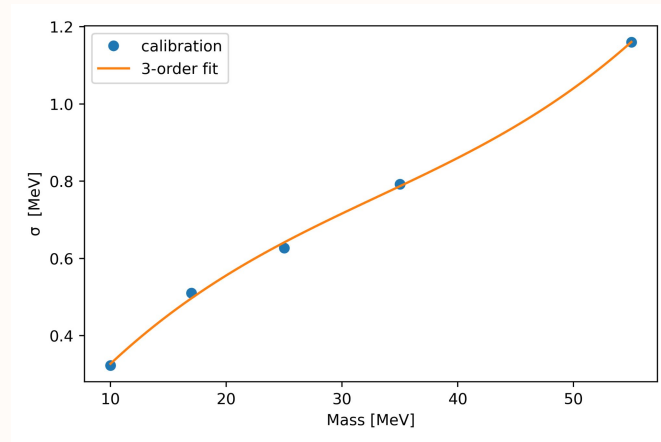
- Bumps appear as excesses with widths of the expected mass resolutions
- Finding these bumps indicate the existence of a particle.
- For the X17, we conduct a prompt resonance search.
- Prompt – looking for the direct production of the X17, not for displaced decays or secondary products.
- Resonance Search – looking for a bump in the observed invariant mass distribution.



The discovery of the Higgs Boson
arXiv:1207.7214

A Bump Hunt, cont.

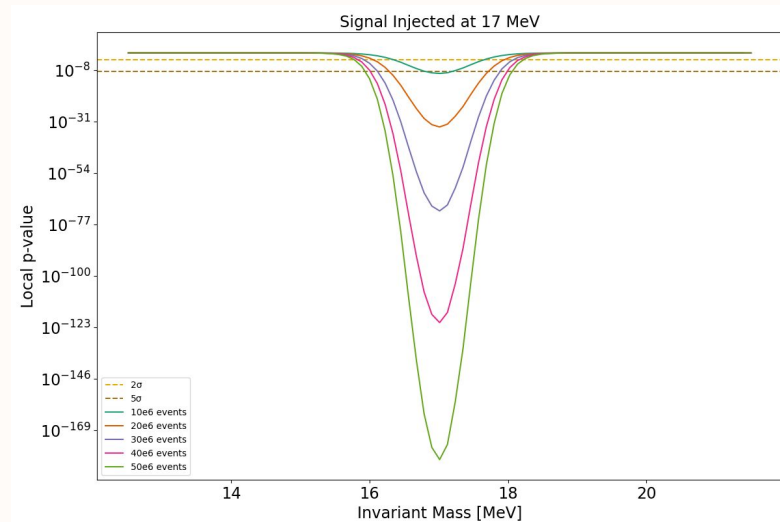
- Given mass resolution values, we generate a formula for the mass resolution over the range we are searching.
- As a proof of concept for the X17 JLab experiment, we can inject various amounts of signal over a background scaled for a 40 day run.



Poisson sampled with additional jitter

A Bump Hunt, cont.

- First, scan over the mass distribution with a window of fixed size (typically on the order of the mass resolution). Calculate the local p-value by comparing the observed counts and expected background counts within the window.
- This plot was generated by pyBumpHunter, which uses the BumpHunter algorithm. (arXiv:1101.0390)
- Next, we need to account for the look-elsewhere effect by adjusting local p-values for the fact that we are conducting so many hypothesis tests.
- The BumpHunter algorithm accomplishes this by comparing the negative log of the minimum local p-value with the significance of pseudo-data generated from the background.



03

Gaussian Process Regression

Problems With Traditional Bump Hunting

Needs a Background Model

Doing polynomial or more complicated fits to data can overfit to excesses caused by particles like X17.

Gaussian Processes

Modelling the background as a gaussian process can lead to more accurate statistical tests.

Gaussian processes also have parameters that can account for uncertainty and sparsity in the data.

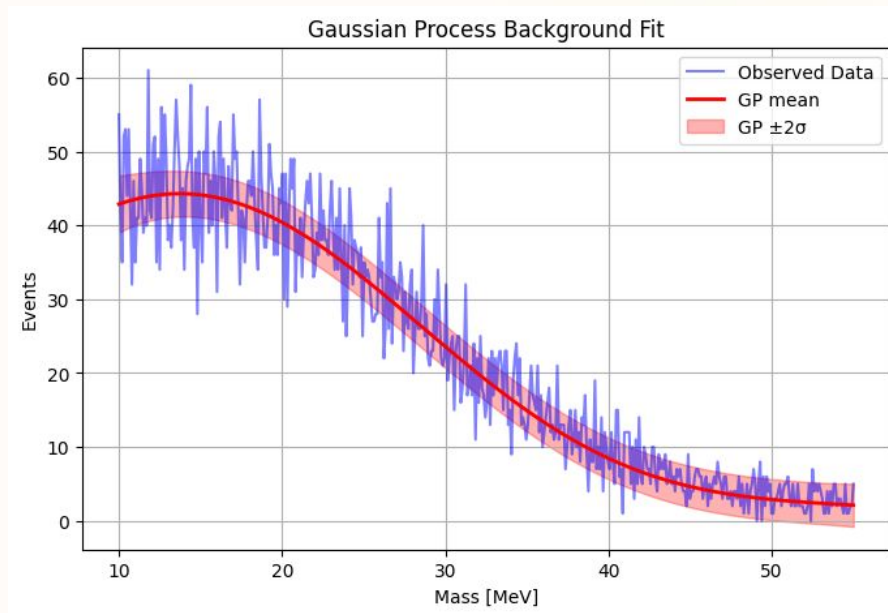
A recent paper (arXiv:2503.07289) applies Gaussian Process Regression to CMS (Compact Muon Solenoid) data.

Gaussian Process Regression

- Here is an example of a gaussian process regression – fitting data to a gaussian process.

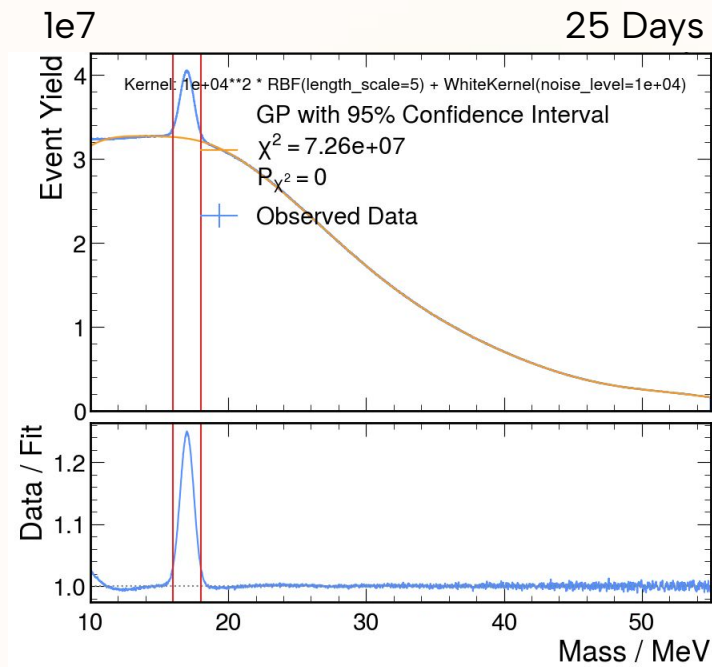
$$p(y_i; \mu_i, \mathbf{K}) = \frac{1}{\sqrt{(2\pi)^n |\mathbf{K}|}} \exp \left(-\frac{1}{2} \sum_{i,j} (y_i - \mu_i) K_{ij}^{-1} (y_j - \mu_j) \right)$$

- y_i is the data points, μ_i is the prior mean, and $\mathbf{K}$ is a covariance matrix parameterized by a specifically chosen kernel function.
- Kernel functions change how data points impact each other affect the smoothness and non-linearity of the GP.
- It is evident that GPs perform well even when data is sparse and noisy by increasing uncertainty at these points.



GP Bump Hunting

- Using a codebase developed by Aidan Hsu, we did a demonstration of GP bump hunting with signal injected at 17 MeV.
- In contrast to pyBumpHunter, this bump hunt didn't need the expected background as input, and learned the background on its own.



- Doing bump hunts with gaussian process regression is a viable alternative to traditional methods that rely on a strong understanding of the background distribution.
- Gaussian processes can also quantify uncertainty in noisy/unreliable data, which can lead to stronger claims about significance and discovery.
- The goal is to develop an easy to use software package for the X17 experiment at JLAB to use once collection begins.
- We can also experiment with different kernels to get the most ideal fit.

CONCLUSIONS & NEXT STEPS

THANKS

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Emrys Peets

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